TRYOLABS IT ADMIN PROVISIONING SPECIFICATION • TELECOM EDITION (TELECOM NEXUS)

Delivered by Tryolabs • Powered by Google Antigravity & Gemini Enterprise

IT Administrator Workstation Requirements — Telecom Edition (Telecom Nexus)

Complete OS, CLI Binary, Python Library (`requirements.txt`), Port, EDR & File Permission Specification for Non-Admin Computers (`antigravity-workshop-telecom`)

CRITICAL REQUIREMENT: ATTENDEES DO NOT HAVE INSTALL PERMISSIONS
Workshop attendees will use locked-down corporate computers without local administrator (**sudo** / Administrator) privileges. IT Administrators must pre-install the system binaries below and pre-build the **`antigravity-workshop-telecom/.venv`** virtual environment using **`admin_provision_workstation.sh`** (macOS/Linux) or **`admin_provision_workstation.ps1`** (Windows), then grant recursive read/write ownership of the **`antigravity-workshop-telecom`** directory to the standard attendee user account.

1. Hardware, OS, Local Loopback Ports & EDR/Firewall Rules

Category	Requirement Specification
Operating System	macOS 13+ (Ventura/Sonoma/Sequoia), Windows 10/11 (64-bit) with Git Bash or WSL2, or Linux (Ubuntu 22.04+, Debian 12+, RHEL 9+)
CPU & Memory (RAM)	Minimum 4 vCPU / 8 GB RAM (8+ vCPU / 16 GB RAM recommended); 10 GB free disk space
Local TCP Ports (Loopback Only)	Telecom Nexus Web NOC UI (**`127.0.0.1:8001`**) and **`127.0.0.1:8080`** (Google ADK Dev UI **`adk web adk_agents`** for **`adk_agents/telecom_nexus`**) — Unprivileged local binding only
EDR / Antivirus Subprocess Rule	Allow **`.venv/bin/python`** to run as a child process over **`stdio`** inside **`antigravity-workshop-telecom/`** for FastMCP (**`mcp_server/telecom_mcp.py`**) and **`PreToolUse`** hooks (**`scripts/audit_hook.py`**)
Outbound HTTPS Allowlist (443)	**`*.googleapis.com`** (**`cloudaicompanion`**, **`aiplatform`**, **`discoveryengine`**, **`run`**), **`accounts.google.com`**, **`oauth2.googleapis.com`**

2. Complete System Software & CLI Binary Matrix (Must Be in System PATH)

Software / Binary	Version	Verify Command	Role in Telecom Edition (Telecom Nexus)	Status
1. Google Antigravity	Latest Enterprise	**`agy --version`** / App Launch	Antigravity 2.0 Desktop App, Antigravity IDE, and **`agy`** CLI in PATH	REQUIRED
2. Python 3 Runtime	>= 3.11 (>= 3.10 ok)	**`python3 --version`**	Must include standard modules **`venv`**, **`pip`**, **`sqlite3`**, **`decimal`**	REQUIRED
3. Astral **`uv`**	>= 0.5.0	**`uv --version`**	Fast Python environment & package manager in system PATH	REQUIRED
4. Git CLI	>= 2.39	**`git --version`**	Cloning **`antigravity-workshop-telecom`** and inspecting diffs	REQUIRED
5. SQLite 3 CLI	>= 3.37	**`sqlite3 --version`**	Lab 1 & Lab 3 database inspection (**`data/telecom_nexus.db`**) and hook tests	REQUIRED
6. Bash & cURL	bash >= 3.2, curl	**`bash --version`**	Executes **`./install.sh`**, **`./verify.sh`**, and local A2A checks	REQUIRED
7. Terraform CLI	>= 1.5.0	**`terraform -version`**	Lab 4 Cloud Run & IAM validation (**`infra/terraform/`**)	REQUIRED (Lab 4)
8. Google Cloud SDK	>= 460.0.0	**`gcloud --version`**	Gemini Enterprise auth & Lab 4 deployment (**`gcloud auth`**)	REQUIRED (Lab 4)
9. Chromium Browser	Chrome / Edge	Open browser	Viewing **`:8001`** dashboard, **`:8080 adk web`**, & HTML slides	REQUIRED

3. Complete Python Library Specification (`antigravity-workshop-telecom/requirements.txt`)

All Python libraries below must be pre-installed by the IT Administrator inside **`antigravity-workshop-telecom/.venv`** so that non-admin attendees never need to run **`pip install`** during the workshop:

Python Package	Min Version	Locked Version	Usage Across Telecom Edition (Telecom Nexus) Labs 1–4
`fastapi`	>=0.110.0	0.141.1	Core REST API & A2A JSON-RPC 2.0 endpoint (**`/api/v1/a2a`**) across Labs 1–4
`uvicorn`	>=0.27.0	0.54.0	Local ASGI server for the live dashboard on **`127.0.0.1:8001`**
`pydantic`	>=2.6.0	2.14.1	Strict schema validation for 3GPP RAN cells, TAP3 CDRs, B2B SLA contracts, TMF622 orders, and A2A payloads

Python Package	Min Version	Locked Version	Usage Across Telecom Edition (Telecom Nexus) Labs 1–4
`pytest`	>=8.0.0	9.1.1	Automated verification suite executed by **`./verify.sh`** across Labs 1, 2, 3, and 4
`httpx`	>=0.27.0	0.28.1	FastAPI **`TestClient`** transport and async A2A protocol requests
`mcp`	>=1.2.0	2.3.0	Official Model Context Protocol SDK (**`FastMCP`**) for Lab 3 (**`mcp_server/telecom_mcp.py`**)
`google-adk`	>=1.0.0	2.12.0	Google Agent Development Kit (**`LlmAgent`**, **`Runner`**, and **`adk web`** for **`adk_agents/telecom_nexus`** on **`:8080`**)
`google-genai`	>=1.0.0	2.33.0	Google GenAI / Vertex AI client SDK used by ADK multi-agent hierarchies
`google-auth`	>=2.28.0	2.62.0	Google Cloud IAM & Application Default Credentials authentication
`aiosqlite`	>=0.20.0	0.22.1	Async SQLite session storage backend required by **`adk web`**
`sse-starlette` & **`starlette`**	>=2.0.0 / >=0.36.0	3.5.0 / 1.9.0	ASGI routing and Server-Sent Events streaming for FastMCP and ADK
`websockets`	>=12.0	15.0.1	Real-time bidirectional event streaming in **`adk web`**
`python-dotenv` & **`PyYAML`**	>=1.0.0 / >=6.0.1	1.2.4 / 6.0.3	Environment configuration and YAML frontmatter parsing for Lab 2 **`SKILL.md`**
`python-multipart` & **`requests`**	>=0.0.9 / >=2.31.0	0.0.32 / 2.34.2	FastAPI form handling and HTTP utilities
`tenacity`, **`watchdog`**, **`graphviz`**	>=8.2 / >=4.0 / >=0.20	9.1.4 / 6.0.0 / 0.21	Agent retry resilience, file watching, and ADK agent graph rendering
`opentelemetry-api` & **`sdk`**	>=1.24.0	1.42.1	Multi-agent execution trace telemetry inspected in **`adk web`**

4. Exact `requirements.txt` for Telecom Edition (Telecom Nexus)

Use the included **`requirements.txt`** (and **`requirements-lock.txt`** for air-gapped wheelhouses) located in **`antigravity-workshop-telecom/requirements.txt`**:

```
# requirements.txt — Tryolabs Google Antigravity Workshop: Telecom Edition (Telecom Nexus)
fastapi>=0.110.0
uvicorn>=0.27.0
starlette>=0.36.0
pydantic>=2.6.0
python-multipart>=0.0.9
python-dotenv>=1.0.0
pytest>=8.0.0
httpx>=0.27.0
anyio>=4.0.0
requests>=2.31.0
mcp>=1.2.0
sse-starlette>=2.0.0
google-adk>=1.0.0
google-genai>=1.0.0
google-auth>=2.28.0
aiosqlite>=0.20.0
websockets>=12.0
PyYAML>=6.0.1
tenacity>=8.2.0
watchdog>=4.0.0
graphviz>=0.20
opentelemetry-api>=1.24.0
opentelemetry-sdk>=1.24.0
```

5. Critical File Permissions for Non-Admin Attendees & One-Step Provisioning

Because attendees run under unprivileged OS user accounts, **the attendee's user account MUST have full read/write/modify permissions** on the entire **`antigravity-workshop-telecom`** folder (**`.venv/`**, **`data/telecom_nexus.db`**, **`.agents/`**, **`src/`**, **`mcp_server/telecom_mcp.py`**, **`adk_agents/telecom_nexus/`**, **`labs/`**). Run our automated provisioning script as Admin to build **`.venv`**, seed **`data/telecom_nexus.db`**, and set user ownership automatically:

```
# Option A: Automated IT Admin Provisioning & Ownership Assignment (macOS / Linux)
cd antigravity-workshop-telecom
sudo ./admin_requirements/admin_provision_workstation.sh --user <attendee_username>

# Option B: Automated IT Admin Provisioning & ACL Assignment (Windows PowerShell as Admin)
cd antigravity-workshop-telecom
.\admin_requirements\admin_provision_workstation.ps1 -AttendeeUser "CORP\<attendee_username>"

# Option C: Verify Readiness as the Unprivileged Attendee User (Zero-Install Mode)
./admin_requirements/admin_provision_workstation.sh --verify-only
```

6. Google Cloud APIs, IAM Roles & Permissions (Antigravity + Gemini Enterprise)

To use **Google Antigravity** (**agy**, Antigravity IDE, and Antigravity 2.0) with **Gemini Enterprise**, execute **Google ADK (`adk web`)** against **Vertex AI**, and deploy/register the **`telecom-nexus`** A2A agent in Lab 4, the Cloud Administrator must enable the APIs and assign the IAM roles below on the workshop Google Cloud project (**`<WORKSHOP_GCP_PROJECT_ID>`**).

6.1 Required Google Cloud APIs to Enable (`gcloud services enable`)

Google Cloud API Service	API Title	Purpose in Telecom Edition (Telecom Nexus)	Status
`cloudaicompanion.googleapis.com`	Gemini for Google Cloud API	Core backend API for Google Antigravity (**`agy`**, IDE, 2.0) under Gemini Enterprise subscriptions, code generation, agentic tool execution, and seat entitlement verification	REQUIRED (All Labs)
`aiplatform.googleapis.com`	Vertex AI API	Powers Google ADK (**`adk web adk_agents`** / **`adk_agents/telecom_nexus`**) and **`google-genai`** SDK (**`GOOGLE_GENAI_USE_VERTEXAI=TRUE`**) for Lab 4 multi-agent execution	REQUIRED (Lab 4)
`discoveryengine.googleapis.com`	Gemini Enterprise / Discovery Engine API	Registers and invokes the **`telecom-nexus`** A2A (**`AgentCard`**) endpoint inside Gemini Enterprise (**`register_gemini_enterprise_agent.sh`**)	REQUIRED (Lab 4)
`serviceusage.googleapis.com` & **`cloudresourcemanager.googleapis.com`**	Service Usage & Resource Manager APIs	Quota project validation (**`x-goog-user-project`**), API state verification, and Terraform provider initialization	REQUIRED (All Labs)
`run.googleapis.com`, **`artifactregistry.googleapis.com`**, **`cloudbuild.googleapis.com`**, **`iam.googleapis.com`**, **`logging.googleapis.com`**	Cloud Run, Artifact Registry, Cloud Build, IAM & Logging APIs	Deploys the containerized **`telecom-nexus`** FastAPI + ADK + A2A service (**`infra/terraform/`**) and records **`cloudaicompanion.googleapis.com`** Cloud Audit Logs	REQUIRED (Lab 4 Deploy)

6.2 Required IAM Roles & Permissions (Attendee Accounts & Cloud Run Runtime)

In addition to assigning an active **Gemini Enterprise / Gemini Code Assist Enterprise license seat** to each attendee in **Google Cloud Console -> Admin for Gemini -> License Management**, grant the following IAM roles:

IAM Role ID	Assigned Principal	Key Included IAM Permissions	Workshop Requirement
`roles/cloudaicompanion.user`	Every Attendee (`user:<attendee>@company.com`)	`cloudaicompanion.companions.generateChat`, `cloudaicompanion.companions.generateCode`, `cloudaicompanion.instances.completeCode`, `cloudaicompanion.instances.generateChat`, `cloudaicompanion.instances.generateCode`, `cloudaicompanion.instances.queryEffectiveSetting`, `cloudaicompanion.entitlements.get`	Mandatory to sign in to Google Antigravity (`agy`, IDE, 2.0) and consume a Gemini Enterprise seat
`roles/serviceusage.serviceUsageConsumer`	Every Attendee (`user:<attendee>@company.com`)	`serviceusage.services.use`, `serviceusage.services.get`, `serviceusage.services.list`	Mandatory to bill API quota calls against the workshop GCP project (`gcloud auth application-default set-quota-project`)
`roles/aiplatform.user`	Every Attendee & `telecom-nexus-sa` Service Account	`aiplatform.endpoints.predict`, `aiplatform.endpoints.generateContent`, `aiplatform.endpoints.streamGenerateContent`, `aiplatform.models.get`	Mandatory for Lab 4 ADK (`adk web adk_agents` / `adk_agents/telecom_nexus`) and multi-agent Vertex AI execution
`roles/discoveryengine.editor` (or `roles/discoveryengine.viewer`)	Attendee / Instructor (Lab 4)	`discoveryengine.engines.get`, `discoveryengine.engines.list`, `discoveryengine.assistants.get`, `discoveryengine.agents.create`, `discoveryengine.agents.update`	Required in Lab 4 to register `telecom-nexus` (`/.well-known/agent.json`) with Gemini Enterprise
`roles/cloudaicompanion.admin` & `roles/logging.privateLogViewer`	Cloud Admin / Workshop Instructor	`cloudaicompanion.entitlements.*`, `cloudaicompanion.instances.*`, `logging.logEntries.list`	Used by Admin to assign Gemini Enterprise license seats and by Instructor to demo Cloud Audit Logs
`roles/run.admin`, `roles/artifactregistry.admin`, `roles/cloudbuild.builds.editor`, `roles/iam.serviceAccountAdmin`, `roles/iam.serviceAccountUser`	Instructor / Terraform Deployer (Lab 4)	`run.services.create`, `artifactregistry.repositories.create`, `cloudbuild.builds.create`, `iam.serviceAccounts.actAs`	Required to run `terraform apply` (`infra/terraform/`) and deploy `telecom-nexus` to Cloud Run v2

6.3 One-Step `gcloud` Commands to Enable APIs & Grant Attendee Roles

```
export PROJECT_ID="<WORKSHOP_GCP_PROJECT_ID>"
gcloud services enable cloudaicompanion.googleapis.com aiplatform.googleapis.com \
  discoveryengine.googleapis.com serviceusage.googleapis.com \
  cloudresourcemanager.googleapis.com run.googleapis.com \
  artifactregistry.googleapis.com cloudbuild.googleapis.com \
  iam.googleapis.com logging.googleapis.com --project="${PROJECT_ID}"

for ROLE in roles/cloudaicompanion.user roles/serviceusage.serviceUsageConsumer \
  roles/aiplatform.user roles/discoveryengine.editor roles/viewer; do
  gcloud projects add-iam-policy-binding "${PROJECT_ID}" \
    --member="user:<ATTENDEE_EMAIL>" --role="${ROLE}"
done
```

7. Pre-Delivery IT Admin Verification Checklist

#	Check	Pass Criteria (Run as Standard Non-Admin Attendee User)
1	System Binaries in PATH	**`python3`**, **`uv`**, **`git`**, **`sqlite3`**, **`bash`**, **`curl`**, **`terraform`**, **`gcloud`**, and **`agy`** resolve in standard user PATH
2	Pre-Built **`.venv`** Ready	**`./.venv/bin/python -c 'import fastapi, uvicorn, pydantic, pytest, httpx, mcp, google.adk, google.genai'`** exits 0
3	ADK Dev UI Binary Ready	**`./.venv/bin/adk --version`** and **`./scripts/run_adk_web.sh --check`** (verifying **`adk_agents/telecom_nexus`**) exit 0
4	Zero-Install Mode Active	Running **`./install.sh`** as standard user skips network downloads (**`[OK] Pre-provisioned .venv detected`**) and seeds **`data/telecom_nexus.db`**
5	Non-Admin Write Access	Standard attendee user can write to **`data/`**, **`.agents/`**, **`src/`**, and **`mcp_server/telecom_mcp.py`** without **`Permission denied`** errors
6	GCP IAM & APIs Ready	**`cloudaicompanion.googleapis.com`** & **`aiplatform.googleapis.com`** enabled; attendee has **`roles/cloudaicompanion.user`** & **`roles/serviceusage.serviceUsageConsumer`**